CAN

PROTOCOL

Features Cont'd

9. CAN Bus speeds & lengths
   - 125kbits/s – 500m
   - 250kbits/s – 200m
   - 500kbits/s – 100m
   - 1Mbits/s – 40m
10. All Messages are BROADCASTED. Any Node is allowed to BROADCAST a message.
11. Each Message contains an ID that identifies the source or content of a message.
12. Each Receiver decides to process or ignore each message , hence message-oriented communication.
13. Is a reliable & robust well proven technology.
14. Support Error detection, Retransmission, & Fault Confinement(temporarily & permanently).

Features Cont'd

15. Inexpensive
16. Bus level standards: CAN 2.0A,CAN2.0B & CAN FD
17. ISO 11898 standards – 6 parts: 1 - 6

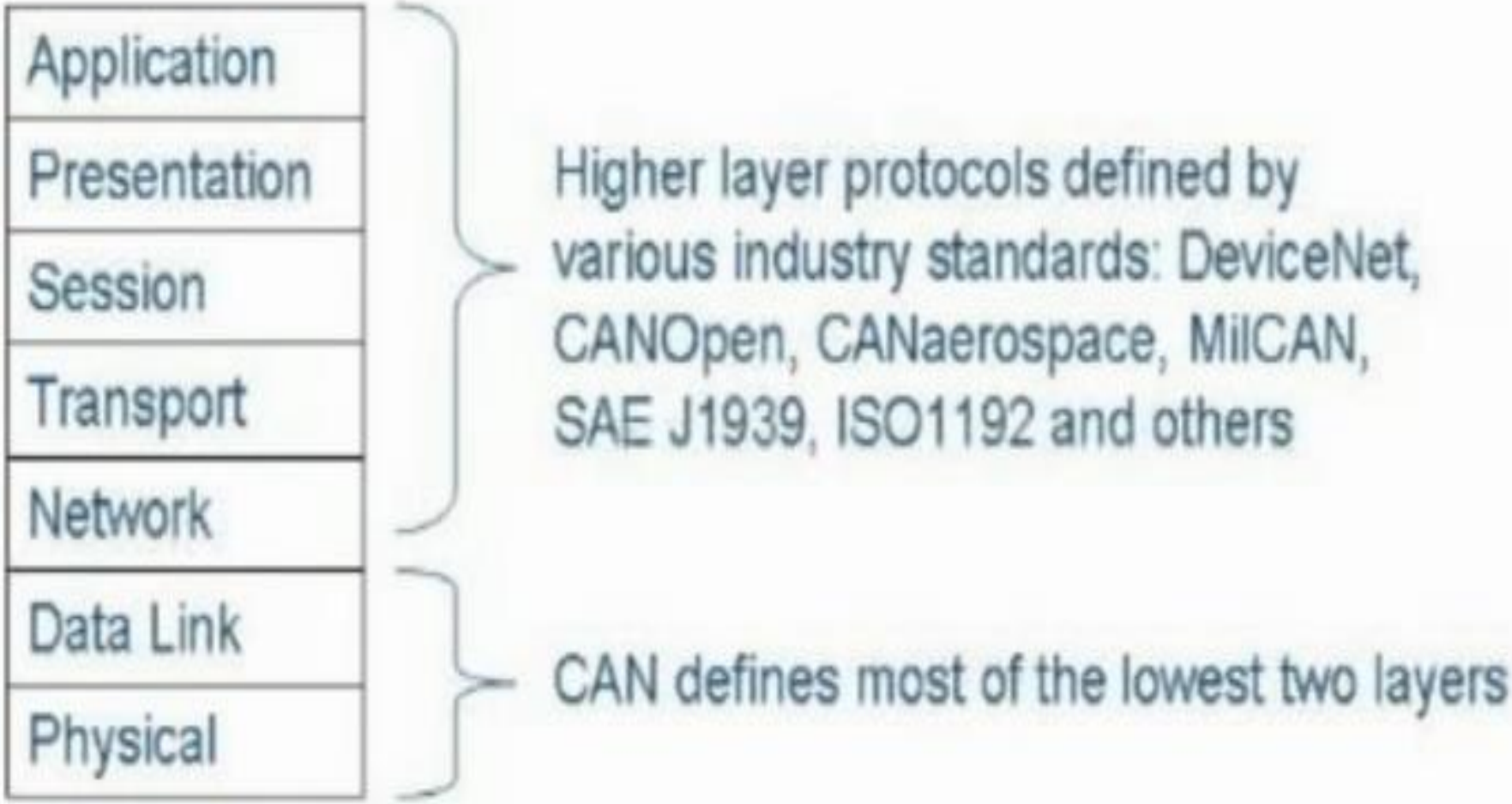
ISO11898 (automotive) CAN standard

- Part 1: Data link layer and physical signaling
- **Part 2: High-speed medium access**
- Part 3: Low-speed, fault-tolerant, medium-dependent interface
- Part 4: Time-triggered communication
- **Part 5: High-speed medium access unit with low-power mode**
  - **Includes all requirements of Part 2, adds low power mode requirements.**
- Part 6: High-speed medium access unit selective wake (***Partial networking***)

Feature's Cont'd

18. CAN bus is one of five protocols used in the **OBD-II/ EOBD** vehicle diagnostics standards.
19. Network Layered Model.

Network Layered Model



Additional CAN-based standards and protocols

HIGHER LAYER PROTOCOLS

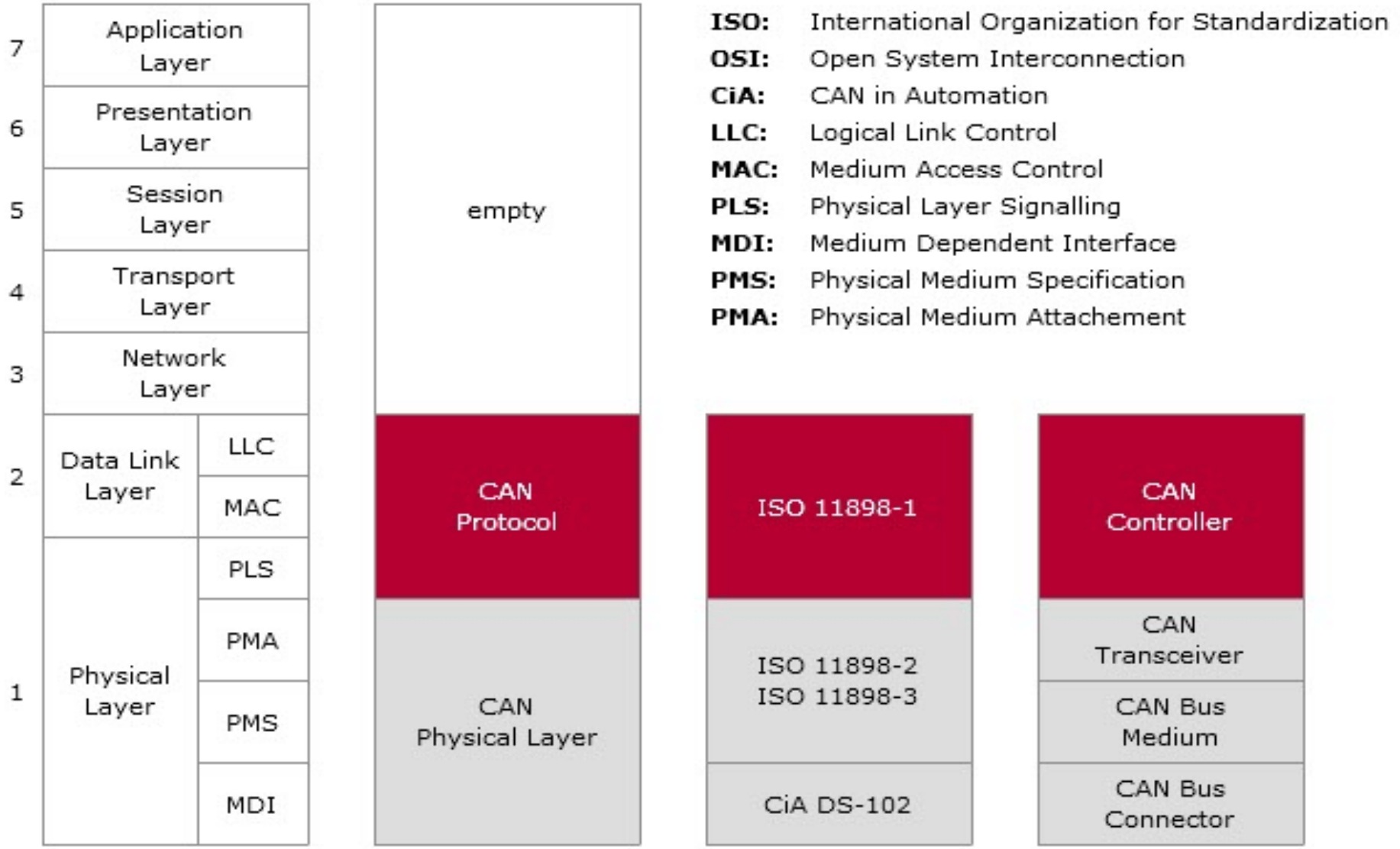
- **ARINC825**: Airborne systems
- **CANaerospace**: Aerospace
- **CAN Kingdom**: Fieldbus
- **CANopen**: Embedded control
- **DeviceNet**: Industrial automation
- **ISO11783**: Agriculture and forestry
- **MilCAN**: Military
- **NMEA2000**: Marine
- **SafetyBUS**: Safety critical automation

ADDITIONAL STANDARDS

- IEC 62132-4
- IEC 61967-4
- ISO 11452
- CISPR 22
- CISPR 25
- EN 55022
- FCC Part 15

CAN Introduction

Standard and Implementation



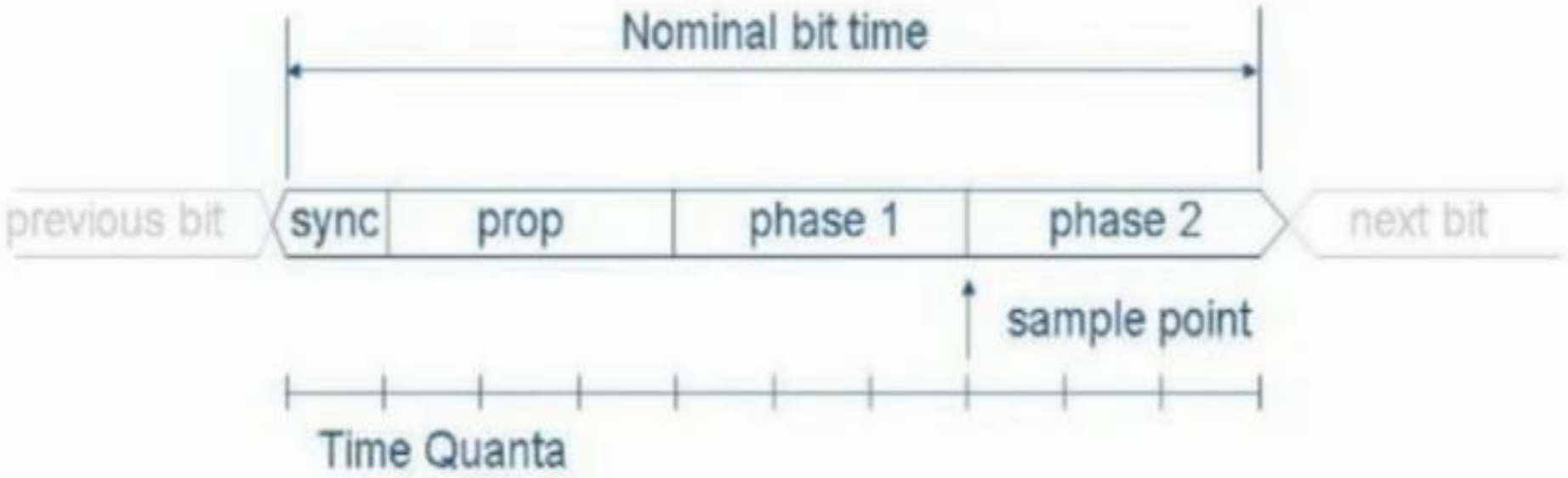
ISO: International Organization for Standardization
OSI: Open System Interconnection
CiA: CAN in Automation
LLC: Logical Link Control
MAC: Medium Access Control
PLS: Physical Layer Signalling
MDI: Medium Dependent Interface
PMS: Physical Medium Specification
PMA: Physical Medium Attachement

20. Four Message Frames/types

- Data Frame – Used to transmit data.
- Remote Frame – Used to request data Transmission.
- Error Frame – Sent by a node that detects an error.
- Overload Frame – sent by node to request a delay in transmission.

21. CAN Bit Time

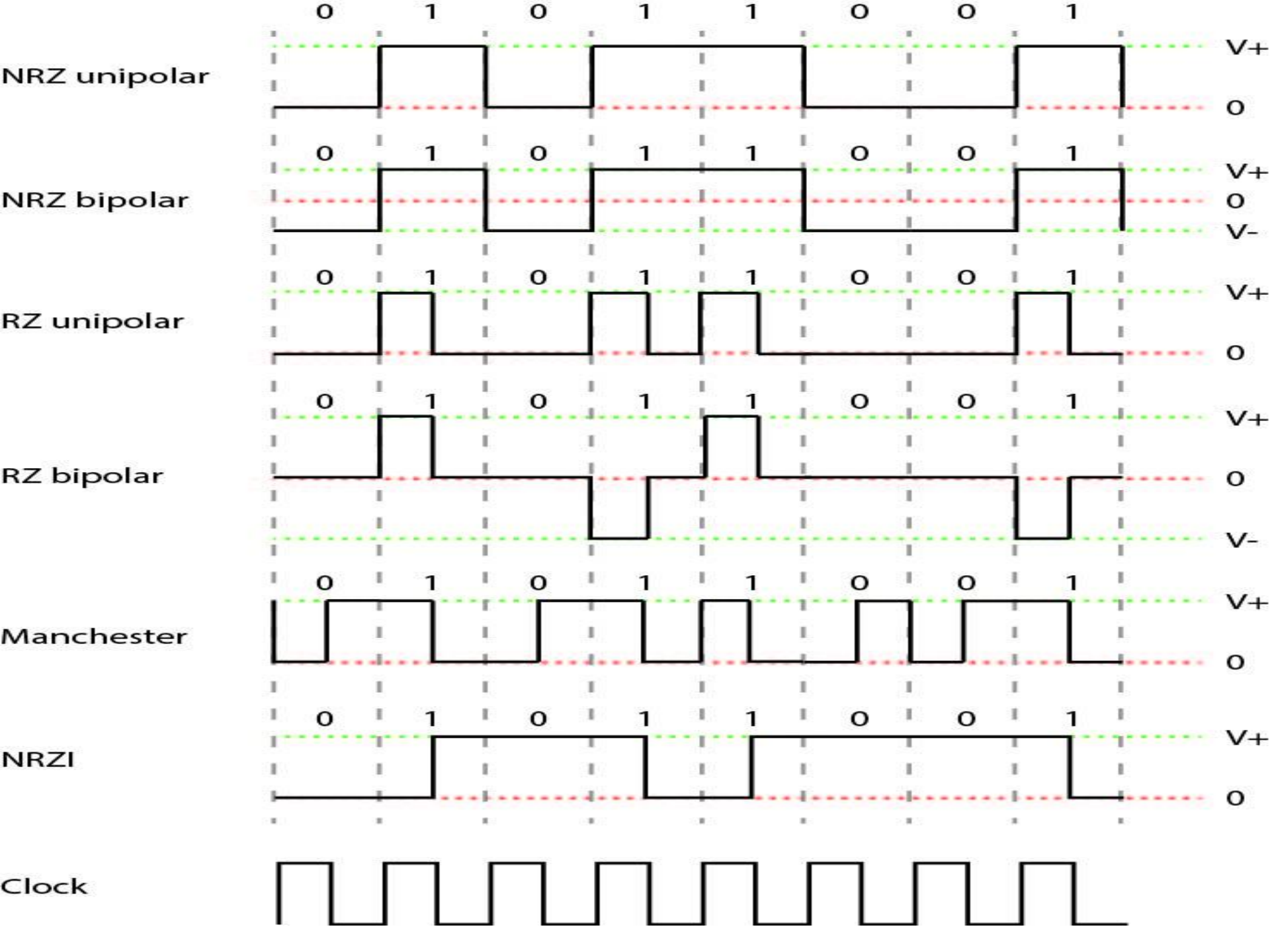
CAN characteristics – Bit timing



Local timing logic re-synchronizes with bus on each recessive to dominant transition

Features Cont’d

22. CAN support Bus Arbitration using wired-and-logic.
23. CAN use NRZ(Non Return to Zero) encoding.


0 1 0 1 1 0 0 1
NRZ unipolar
V+
0
NRZ bipolar
V-
RZ unipolar
RZ bipolar
Manchester
NRZI
Clock

Features Cont'd

24. Error Detection capabilities :

- **Bit error,**
- **Ack Error,**
- **Form Error,**
- **Stuff Error**
- **CRC Error**

25. Reasons For Popularity :

- **low cost – less wiring**
- **Centralized -**
- **Robust – reliable & fault tolerant**
- **Efficient – high rates**
- **Flexible -**

Features Cont’d

25. CAN Network Size
    - The maximum number of nodes is not specified.
    - Networks are limited by electrical loading, up to 64 nodes is normal.

Automotive Body Network Example Architecture

